DMBI CA2 Mini Project · D15A Group 09 · SDG 3 and SDG 11

# BlackSpot Miner

Mining road collision data to locate severe-crash hotspots and prioritise road safety action

**513,801**
real collisions, 2021 to 2025

**804**
street-scale hotspots of fatal and serious crashes

**107 of 109**
rules held on unseen 2025

**0.764**
next-year site ROC-AUC, past counts 0.680

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# A safety budget cannot fix every road, so where should it go?

**1.19 million**

road deaths a year worldwide (WHO, 2023)

**1.72 lakh**

road deaths in India in 2023 (MoRTH)

**SDG 3.6**

target on road traffic deaths and injuries

**Where?**

Which street-scale places concentrate severe collisions?
DBSCAN hotspots

**Why?**

Which conditions raise the chance of a death or serious injury (KSI)?
Apriori rules, K-Modes segments

**What next?**

Which sites should be treated first next year?
Classification and site ranking

# Eight stages take raw records to a decision

**1**

## Problem

Limited budget, where to act

**2**

## Data

513,801 collisions

**3**

## Preprocessing

Audit, clean, encode

**4**

## Algorithm

DBSCAN, Apriori, K-Modes, classifiers

**5**

## Results

Hotspots, rules, site scores

**6**

## Evaluation

Unseen 2025 test year

**7**

## Insights

Places and combinations

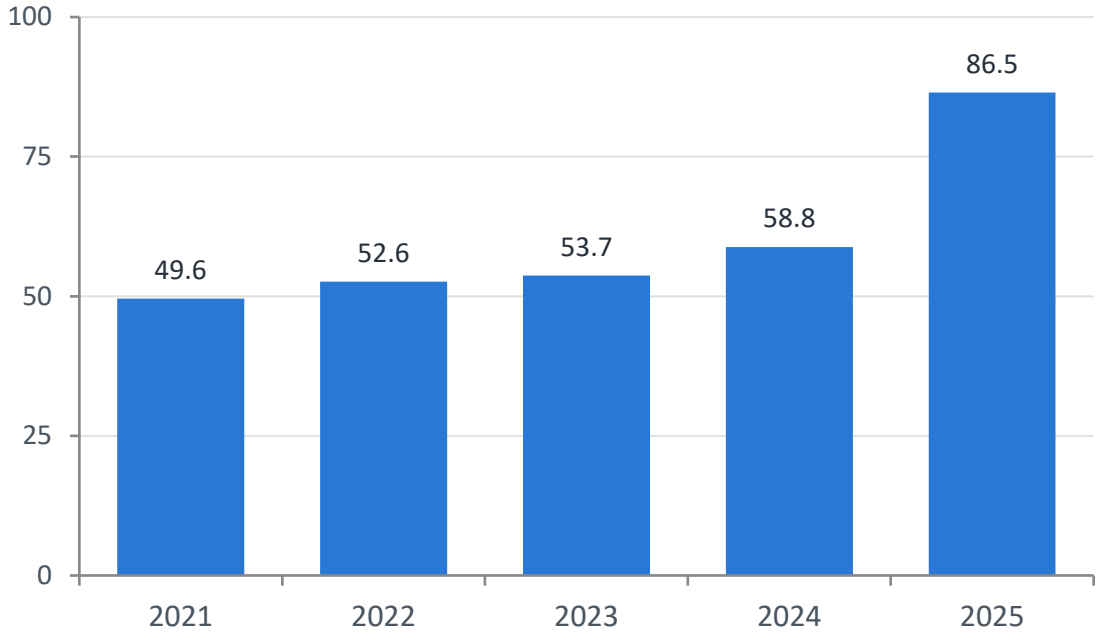
**8**

## Decision

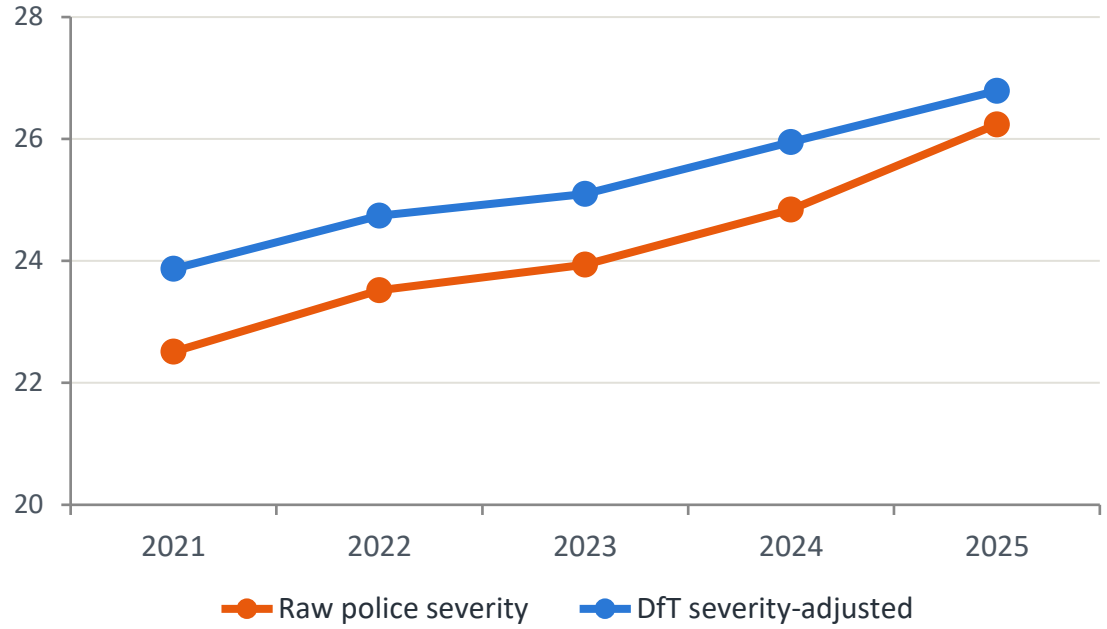
2026 priority list

# A reporting change would have faked a trend, so we audited first

Collisions under injury-based reporting (%)




KSI share of all collisions (%)




## Why it matters

15 of 44 police force codes changed system in five years. Raw KSI share goes from 22.51% to 26.24%. A model trained on it would partly learn a change of form, not a change in road risk.

# Preprocessing made five years of records comparable

### Integrate

Join 3 tables on collision_index. Derive flags: motorcycle, pedestrian hit, left the road, overturned.

### Remove leakage

Drop number of casualties, police attendance and enhanced severity. All are known only after the outcome.

### Clean

Remove 56 unusable rows. Drop fields that are 86.5% missing in 2025. Keep Unknown explicit.

### Split by time

Train 221,193 rows on 2021 to 2024. Test 87,848 rows on 2025. No random split.

**513,745**

rows after cleaning, for location ranking

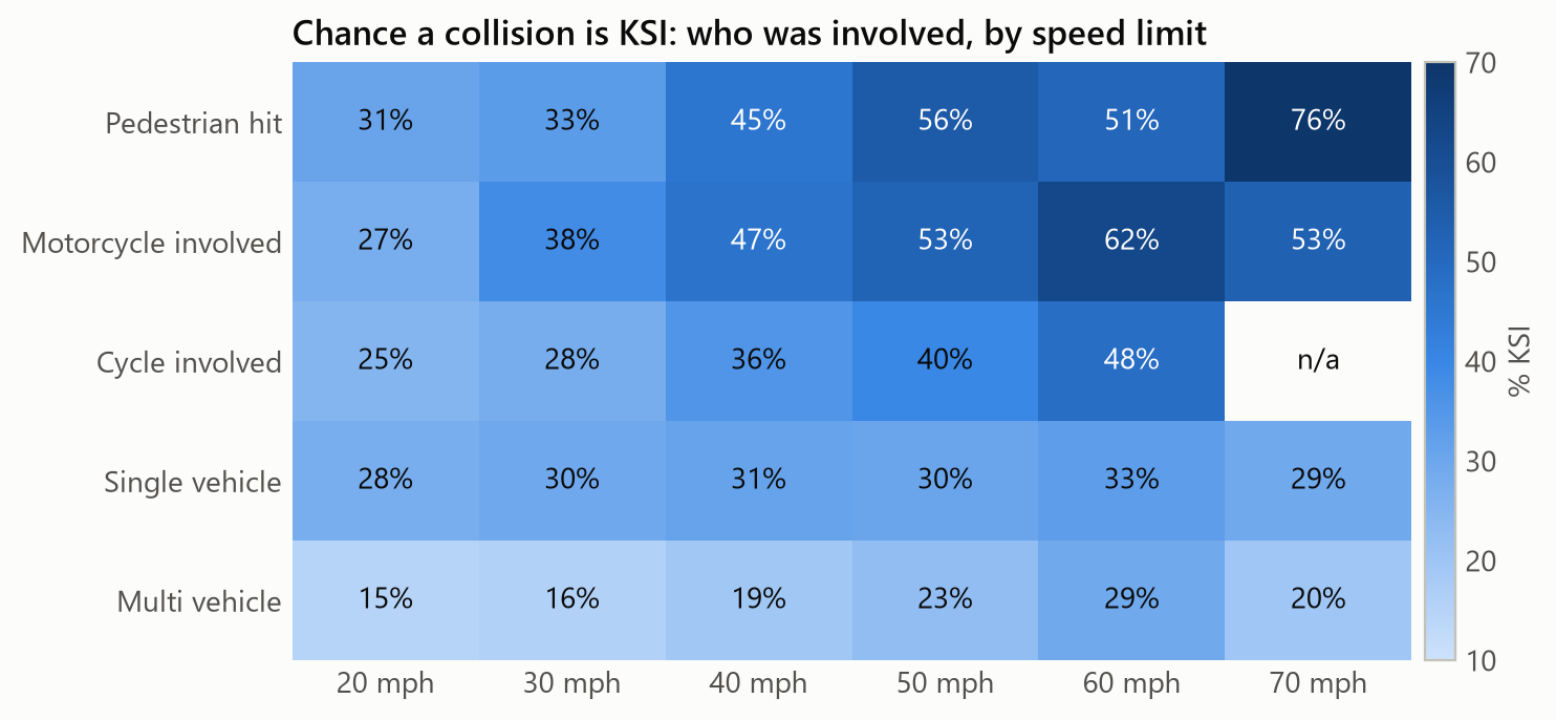
**309,041**

injury-based rows, for rules and severity

**27.45%**

of injury-based collisions are KSI

# Speed hurts vulnerable road users far more than other crashes




Cells with under 200 collisions show n/a. Injury-based records only.

**Most KSI collisions: Fri 15:00. Highest chance of KSI: Mon 04:00 (45.6%).**

**43.6%**

of motorcycle collisions are KSI, against 27.4% overall

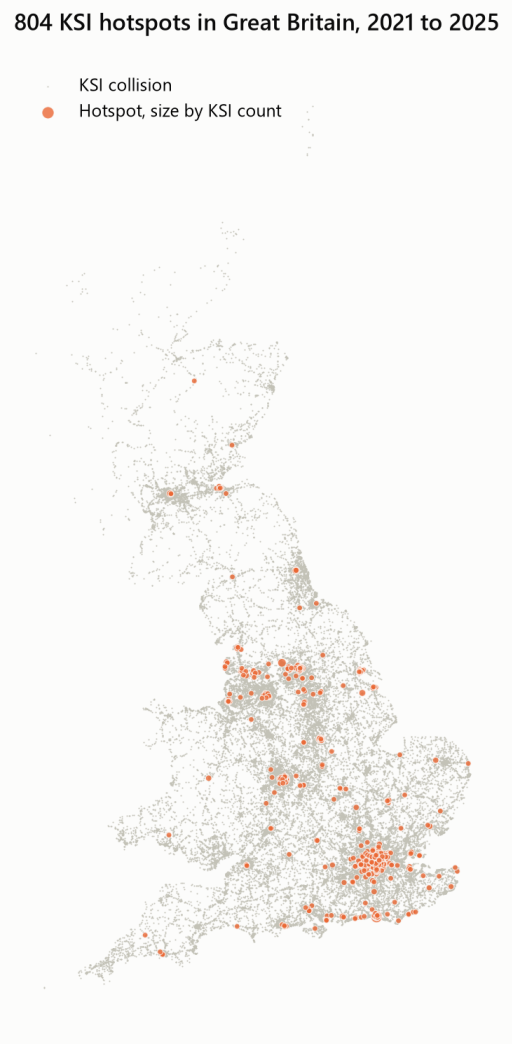
**37.1%**

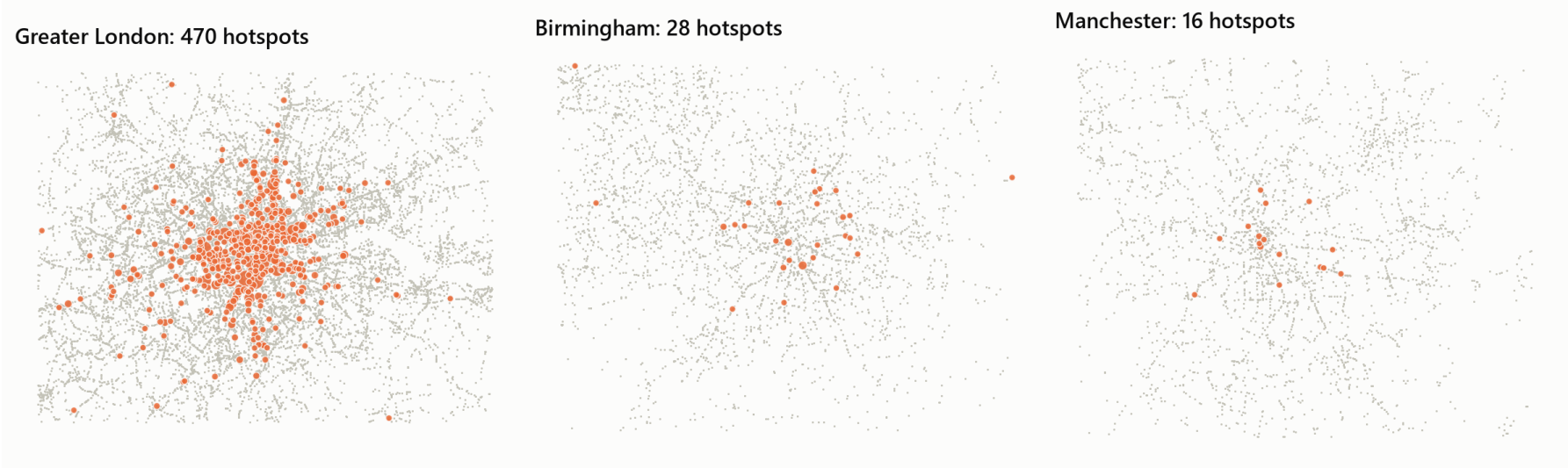
on 60 mph roads; motorways are 25.5%

**36.9%**

in darkness with no lighting; daylight is 26.3%

# DBSCAN found 804 real hotspots, but most severe crashes are scattered




**173 m**

median hotspot span: a place an engineer can visit

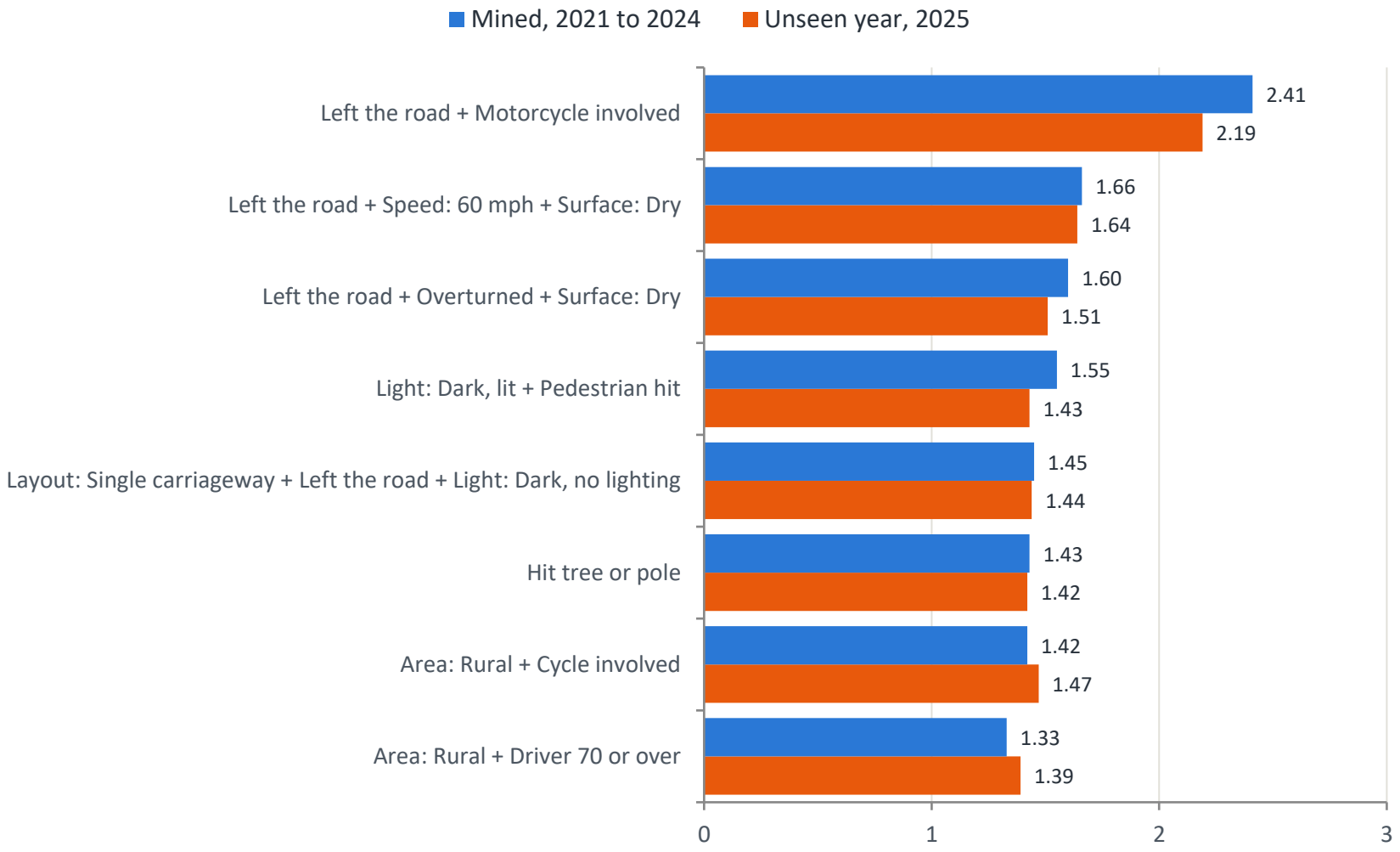
**4.79×**

2025 KSI per km², hotspots against random past sites

**93.89%**

of KSI collisions fall outside any hotspot

# 107 of 109 rules held on a year the mining never saw




**0.946**

correlation of lift between the two periods

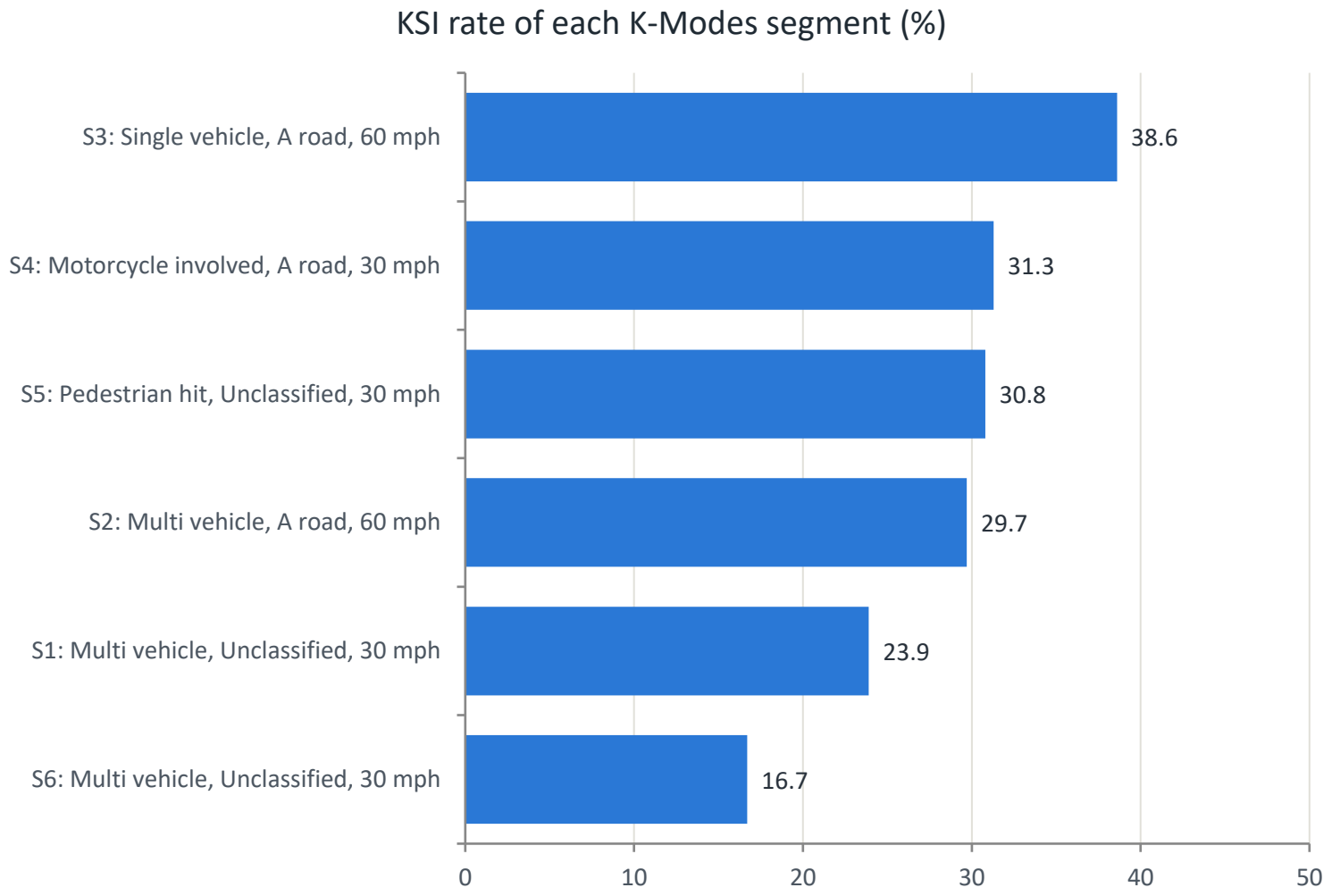
**0.9%**

median fall in lift on unseen data

**Reading lift**

2.0 means KSI is twice as likely as average. It shows association, not cause.

# K-Modes segments confirm our reference paper's claim at scale




**60**

rules found only inside a segment

**55**

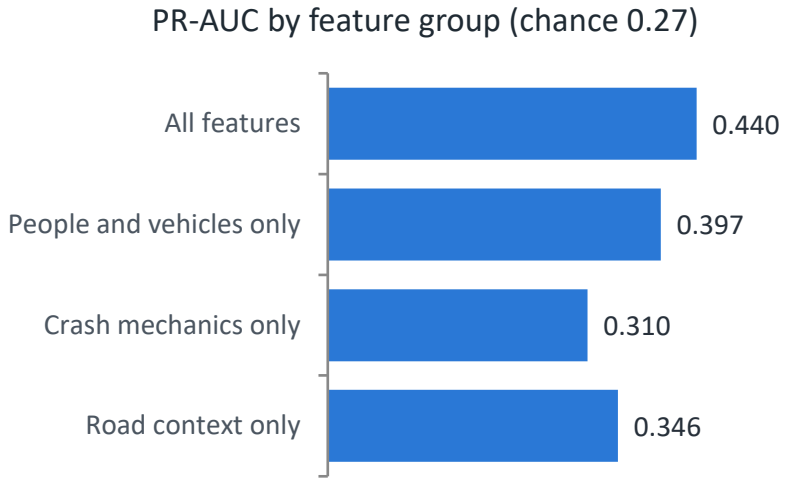
of those held on unseen 2025

**Top hidden rule**

Light: Dark, no lighting + Pedestrian hit: lift 1.78, 68.9% KSI inside S3.

# Severity is only partly predictable, and that is a real finding

| Model | Accuracy | Precision | Recall | F1 | ROC-AUC | PR-AUC |
|---|---|---|---|---|---|---|
| **Random Forest** | **56.9%** | **35.7%** | **74.4%** | **0.482** | **0.679** | **0.440** |
| AdaBoost | 56.9% | 35.4% | 72.3% | 0.475 | 0.670 | 0.426 |
| Decision Tree | 54.7% | 34.5% | 75.6% | 0.474 | 0.665 | 0.422 |
| Naive Bayes | 54.4% | 33.8% | 71.7% | 0.460 | 0.638 | 0.376 |




**73.0%**

accuracy from predicting Slight for everyone, catching no KSI collision

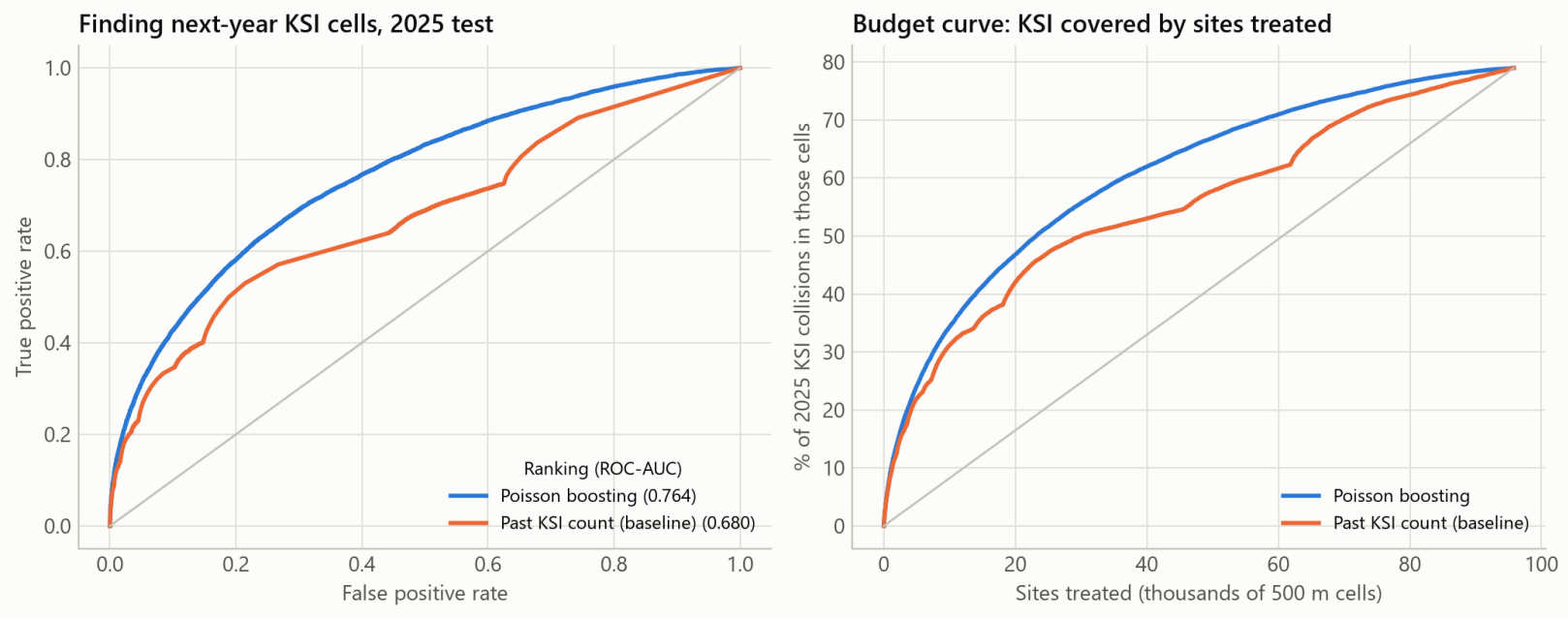
**74.4%**

of KSI collisions caught by Random Forest at the threshold set on 2024

### Why not higher?

Impact speed, seat belt use and the health of the person hit are not recorded. Road context alone adds little.

# Ranking next year's sites beats a past-count list




| Sites treated | Past counts | Poisson boosting |
|---|---|---|
| 500 | 5.4% | 5.8% |
| 1,000 | 8.6% | 9.3% |
| 5,000 | 22.0% | 24.1% |
| 10,000 | 31.2% | 34.3% |

**+2.13 pts**

KSI covered at 5,000 sites (95% interval 1.78 to 2.53)

**-39.8%**

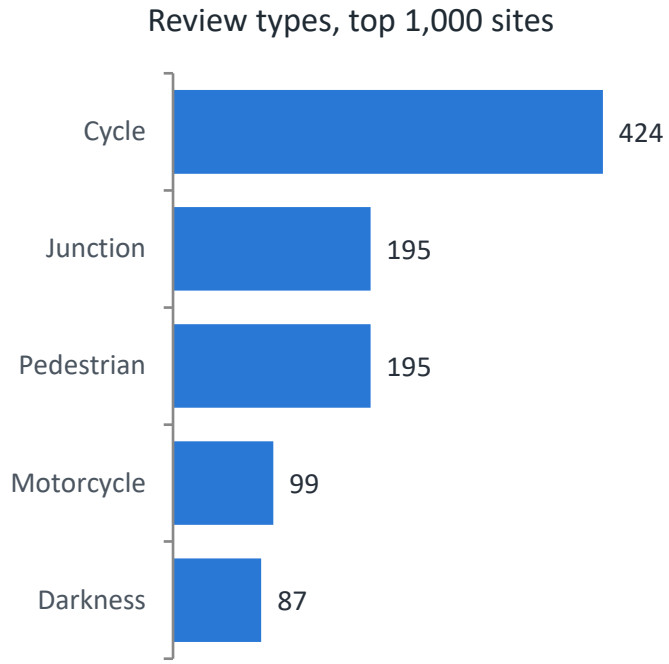
next-year fall in the 100 worst cells of 2024; the data records no treatment

**78.98%**

of 2025 KSI collisions happened in a cell with past collisions. The rest cannot be ranked by history.

# The output is a 2026 priority list an engineer can act on

| # | Road | Authority | KSI 2023-25 | Suggested review |
|---|---|---|---|---|
| 1 | A3 | City of London | 23 | Cycle separation and junction protection |
| 2 | A201 | City of London | 28 | Cycle separation and junction protection |
| 3 | A201 | Southwark | 19 | Cycle separation and junction protection |
| 4 | A3 | Southwark | 19 | Cycle separation and junction protection |
| 5 | A401 | Westminster | 35 | Pedestrian crossing and speed |
| 6 | A40 | Westminster | 21 | Cycle separation and junction protection |
| 7 | A40 | Westminster | 29 | Cycle separation and junction protection |




### How an authority uses it

Choose a budget, take that many sites from the list, read each site's pattern on the interactive map, and send an engineer to verify it.

**98%**

of the top 100 sites are near a DBSCAN hotspot

# What we can claim, and what we cannot

### Tested on unseen 2025

- 804 street-scale hotspots
- 107 of 109 rules held
- Site ranking beats past counts by 2.13 points at 5,000 sites
- The paper's clustering claim holds at scale

### Limits

- No traffic volumes, so busy places rank high
- Association, not cause; no crash reduction claimed
- Police-reported injury collisions only
- Numbers describe Britain; the method transfers to India through eDAR

**SDG 3.6** road deaths and injuries. **SDG 11** sustainable cities and safe transport.

Reference paper: Kumar, S. and Toshniwal, D. (2015). A data mining framework to analyze road accident data. Journal of Big Data, 2, 26.
link.springer.com/article/10.1186/s40537-015-0035-y
Data: Department for Transport, Road Safety Open Data, Open Government Licence v3.0.

## Thank you. We are open for your questions.